Primary and Secondary Amenorrhea

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Block: Endocrine/Reproductive/Urologic

Thread: Clinical Medicine

Conflict of Interest: Dr. Cullen has no conflicts of interest to disclose

Please note that any charts/pictures that include PCOS have not yet been updated to the new terminology, PMOS

Learning Objectives

By exam time, you should be able to do the following:

1. Define primary amenorrhea and recognize its clinical presentation
2. Recognize the differential diagnosis of primary amenorrhea
3. Describe the work-up and treatment considerations for primary amenorrhea
4. Define secondary amenorrhea and recognize its clinical presentation
5. Recognize the differential diagnosis of secondary amenorrhea
6. Describe the work-up and treatment of secondary amenorrhea

Study Questions

1) A 15yo patient presents to her pediatrician with her parent to discuss not having had a first menstrual period. She denies medical or surgical history and takes no medications. She reports growth and development otherwise similar to her peers. Her vital signs are normal and her BMI is 23. Her physical exam reveals typical breast and pubic hair development. On pelvic exam, she is found to have a shortened vagina with no visible or palpable cervix. You are unable to palpate a uterus on exam, but it is difficult to fully evaluate due to patient discomfort. What is currently on your differential and what would your next steps in evaluation be?

2) A 27yo G1P0010 presents to her gynecologist after not having a period for 4 months. Prior to this, her periods had been regular and monthly. She denies past medical history. Her surgical history is significant for an appendectomy at age 12yo. She takes ibuprofen as needed for headaches, but no prescribed medications. She reports menarche at age 12y, initially irregular for the first 3-4 years, then regular for the last 10+ years. This is the longest she has ever gone without a period. She reports weight gain of about 10-15 pounds over the last year. She denies change in activity level or diet. She does note increasing fatigue and that she has had some acne around the time of her periods for the last two years. She does report constipation, but denies nipple discharge, hot flashes or vaginal concerns. She is sexually active with a male partner and uses condoms almost every time. Her VS are normal, BMI is 29, and general PE and pelvic exam are unremarkable. What is on your differential diagnosis and what are your next steps in assessment?

Primary Amenorrhea

Definition

- No periods by age 15y in the presence of normal growth and secondary sexual development
- We should initiate workup for any of the following:
    - No period by age 15y
    - No period PLUS no breast development by age 13y
    - No period within 5y of onset of breast development
    - No period 14y with clinical evidence of hyperandrogenism (hirsutism, cystic acne)

Differential Diagnosis

1. Structural/outflow tract abnormalities
    - 5a-reductase deficiency
        - 46,XY with normal testosterone production but defective conversion of testosterone to dihydrotestosterone (DHT) --> impaired virilization during embryogenesis
        - 5-alpha-reductase type 2 enzyme is defective (various **autosomal recessive** loss of function mutations) but type 1 enzyme is normal
        - Patients may have low (but measurable) serum DHT levels due either to residual activity of the defective enzyme or the normal type 1 isoenzyme
        - Virilization at the time of expected puberty may occur due to increase in serum testosterone or increase in DHT concentration through the Type 1 enzyme
        - Testosterone and estrogen production are normal/expected range for an XY individual, which is why gynecomastia does not develop (as opposed to androgen insensitivity- see next slide).
        - Genital development
            a. External genitalia varies from isolated microphallus with varying degrees of hypospadias (33%), to typical appearing labia and clitoris (7%), labia with clitoromegaly (49%)
            b. The internal urogenital tract consists of epididymis, vas deferens, seminal vesicles, and ejaculatory ducts. These empty into a blind-ending vagina (55%), on the perineum near the urethra (40%), or at the end of the phallus (5%). The testes are outside the abdominal cavity in the inguinal canals, labia majora, or scrotum. Spermatogenesis is most often impaired.
    - Complete Androgen insensitivity syndrome
        - 46 XY with loss of function mutations in the gene that codes for the androgen receptor
        - Complete AIS is the 3rd most common cause of primary amenorrhea
        - Partial AIS has a variable phenotype
        - Little or no axillary or pubic hair
        - In infancy, inguinal mass may be undescended testicle

        a. Testes may be in the abdomen, inguinal canals or labia majora
    - Presence of labia and clitoris, absent uterus, short vagina
    - Elevated LH due to resistance to feedback of androgens at the hypothalamic-pituitary level --> increased testosterone levels
    - Breast development occurs mediated by excess testosterone aromatizing to estrogen
- Mullerian agenesis (aka Vaginal agenesis, Mayer-Rokitansky-Kuster-Hauser syndrome, MRKH)
    - 46 XX karyotype with agenesis or hypoplasia of mullerian duct system
    - Underlying etiology largely unknown/unidentified
    - Congenital absence of the vagina most often accompanied by cervical and uterine agenesis (<10% have an obstructed or rudimentary uterus)
    - 25-50% have urologic abnormalities (horseshoe kidney, unilateral renal agenesis)
    - 10-15% have skeletal abnormalities
    - Normal ovaries and normal ovarian function, thus normal secondary sexual differentiation EXCEPT no menarche (primary amenorrhea)
- Imperforate hymen
    - Hymen is ALWAYS at the vaginal opening, not high in the vagina
    - Hymen is thin: 1-2mm
    - Can appear blue hued when blood trapped behind (hematocolpos)
- Transverse vaginal septum
    - Transverse vaginal septum most commonly 2/3 of the way up the vagina (high septum)
    - Can also be mid or low
    - Septum is thick: 1-2cm



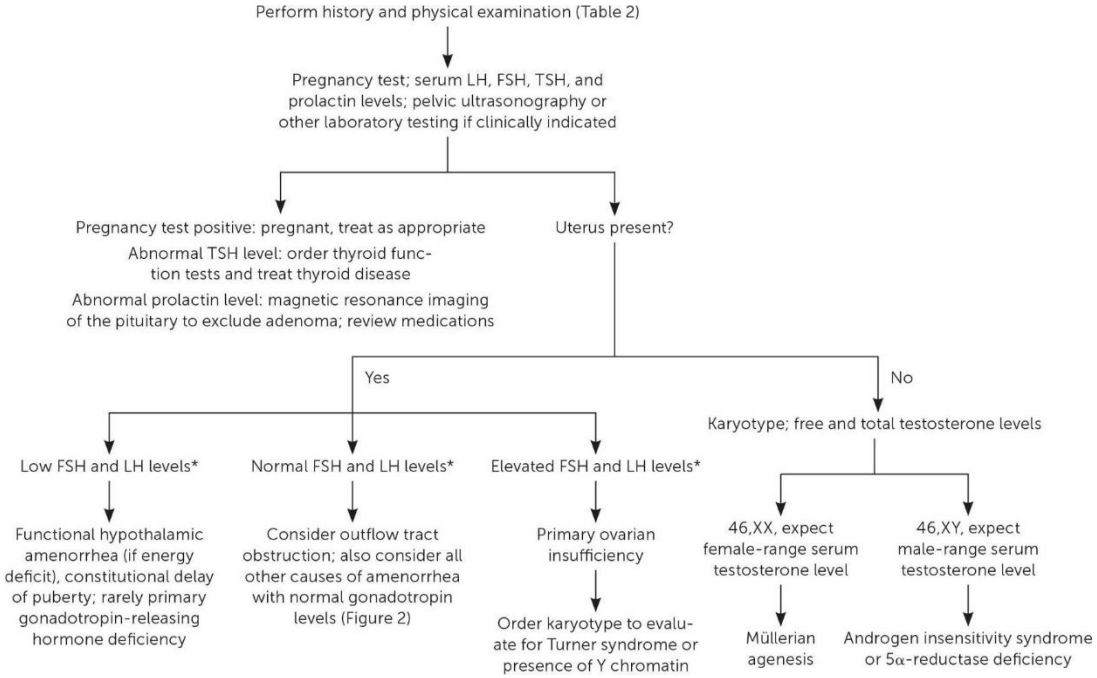
2. Primary ovarian insufficiency
    - Turner syndrome (45X)
        - 45X gonadal dysgenesis- missing an entire X chromosome
        - Can also have mosaic 45X/46XX- spontaneous puberty and menstruation typically occur, followed by secondary amenorrhea
        - Premature depletion of oocytes/follicles (largely in utero) --> decreased estradiol and inhibin B --> significantly elevated FSH due to reduced negative feedback
        - Ovaries are replaced with fibrous tissue
        - Should have estradiol and progesterone therapy
        - Other typical features: short stature, shield chest/widely spaced nipples, short and webbed neck. Hearing loss, hypothyroidism and LFT abnormalities are common with increasing age. Increased risk of cardiovascular disease due to cardiac malformations
    - Other gonadal dysgenesis (46XX or 46XY)
        - 46XX gonadal dysgenesis
            a. Many genetic causes discovered with next generation sequencing

        b. Typically homozygous or compound heterozygous mutations in genes that code for proteins involved in homologous recombination in meiosis and DNA damage/repair
    - 46XY gonadal dysgenesis
        a. Fibrous streak gonad doesn't secrete anti-mullerian hormone (AMH) resulting in persistent mullerian structures
        b. Various mutations have been identified, including mutations in or absence of SRY
  - Idiopathic: Autoimmune mediated?
  - Acquired
    - chemo or radiation exposure early in life
    - Surgical removal due to ovarian tumors
3. Hypothalamic or pituitary disorders
  - GnRH deficiency/defect (aka isolated GnRH deficiency, idiopathic hypogonadotropic hypogonadism)
    - Defects in production and/or action of GnRH
        a. About 1/3 are inherited (autosomal dominant, autosomal recessive, or X-linked recessive inheritance)- many different genes are implicated in etiology.
        b. The remaining 2/3 are sporadic
    - Can occur with normal olfaction or with anosmia (Kallman syndrome)
    - Inappropriately low or low normal FSH and LH in the setting of prepubertal (low) gonadal steroid levels
    - Difficult to directly diagnose with GnRH levels so diagnosis inferred by:
        a. Lack of LH pulses in frequent blood sampling
        b. Robust gonadotrophin secretion in response to pulsatile GnRH administration demonstrating functional response of pituitary
  - Constitutional delay
    - Diagnosis of exclusion
    - Will go on to typical pubertal development, just later
    - Can be difficult to distinguish from GnRH deficiency
    - Genetic basis poorly understood
4. Many of the secondary amenorrhea causes, including pregnancy, thyroid disease, functional amenorrhea, chronic illness, and PMOS can also cause primary amenorrhea. Those will be covered in the next section.

Primary Amenorrhea Work-up: H&P, imaging, laboratory

Findings in the Evaluation of Amenorrhea

Findings	Associations
History	
Chemotherapy or radiation	Impairment of specific organ or structure, (e.g., brain, pituitary, ovary)
Family history of early or delayed menarche	Constitutional delay of puberty
Galactorrhea	Pituitary tumor
Hirsutism, acne	Hyperandrogenism, PCOS, ovarian or adrenal tumor, CAH, Cushing syndrome
Illicit or prescription drug use	Multiple associations, consider effect on prolactin
Loss of smell (anosmia)	Kallman syndrome (GnRH deficiency)
Menarche and menstrual history	Primary vs. secondary amenorrhea
Sexual activity	Pregnancy
Significant headaches or vision changes	Central nervous system tumor, empty sella syndrome
Temperature intolerance, palpitations, diarrhea, constipation, tremor, depression, skin changes	Thyroid disease
Vasomotor symptoms (e.g., hot flashes or night sweats)	Primary ovarian insufficiency, natural menopause
Weight loss, excessive exercise, poor nutrition, psychosocial distress, diets	Functional hypothalamic amenorrhea
Physical examination	
Abnormal thyroid examination	Thyroid disorder
Acanthosis nigricans or skin tags	Hyperinsulinemia (PCOS)
Anthropomorphic measurements; growth charts	Multiple associations; Turner syndrome, constitutional delay of puberty
Body mass index	High: PCOS
	Low: Functional hypothalamic amenorrhea
Bradycardia	Functional hypothalamic amenorrhea (e.g., anorexia nervosa)
Breast development (normal progression)	Presence of circulating estrogen*
Dysmorphic features (e.g., webbed neck, short stature, low hairline)	Turner syndrome
Male pattern baldness, increased facial hair, acne	Hyperandrogenism, PCOS, ovarian or adrenal tumor, CAH, Cushing syndrome
Pelvic examination	
Absence or abnormalities of cervix or uterus	Rare congenital causes including Müllerian agenesis or androgen insensitivity syndrome
Clitoromegaly	Androgen-secreting tumor; CAH; 5α-reductase deficiency
Presence of transverse septum or imperforate hymen	Outflow tract obstruction
Reddened or thin vaginal mucosa	Decreased endogenous estrogen
Sexual maturity rating abnormal	Turner syndrome, constitutional delay of puberty, rare causes
Striae, buffalo hump, central obesity, hypertension	Cushing syndrome



Primary Amenorrhea Treatment

- Diagnosis is the goal, not necessarily periods
    - Risk assessment (undescended testes, cardiovascular disease, etc)
    - Fertility goals/possibilities
    - Addressing social and developmental outcomes
- If due to outflow tract abnormality
    - Obstruction: Remove obstructing tissue
    - Absent uterus/vagina: Assess patient goals
- For secondary causes, treat the cause

Secondary Amenorrhea

Definition

- The absence of menses for six or more months

OR

- The absence of menses for the length of three cycles after the establishment of regular menstrual cycles (e.g. A person with regular 28-30 day cycles would meet this after 3 months)

Differential Diagnosis

1. **Pregnancy**
2. Uterine: Intrauterine adhesions (Asherman Syndrome)
    - Acquired scarring of endometrium
    - Usually secondary to postpartum hemorrhage management or D&C (dilation and curettage) after pregnancy loss or termination of pregnancy
    - Patients must have a preceding event after which their periods became absent or very light- does not happen spontaneously
3. Ovarian
    - Premature Ovarian Insufficiency/Failure
        - Depletion of oocytes under age 40y
        - Low estradiol and inhibin B leads to loss of negative feedback on the hypothalamus/pituitary resulting in high FSH levels
        - Can be seen in mosaic 45X/46XX, fragile X premutation, autoimmune ovarian destruction, chemo/radiation, or undefined causes
    - Ovarian Tumors
        - Rare Inhibin B secreting tumor --> FSH suppression --> estradiol suppression

4. Polyendocrine Metabolic Ovarian Syndrome (PMOS)/Hyperandrogenism
    - PMOS
        - Accounts for 20% of cases of amenorrhea (50% of oligomenorrhea)
        - Clinical Diagnosis
        - Rotterdam Criteria: must have two of the following three
            1. Ovulatory dysfunction
            2. Hyperandrogenism- clinical or laboratory
            3. Polycystic ovaries
        - Lab testing in PMOS:
            - Total testosterone + SHBG **OR** Total + Free testosterone
                a. Serum SHBG may be low --> increased free testosterone
                b. Low level of SHBG is a serum biomarker for insulin resistance, hyperandrogenism, and abnormal glucose and lipid metabolism. SHBG polymorphisms are correlated with PCOS risk
            - Gonadotropin testing may reveal and elevated LH:FSH ratio. LH is often 2-3 times higher than FSH.
            - Insulin resistance is very common in PMOS but is not part of the diagnostic criteria. Nonetheless, patients should have testing for pre-diabetes or diabetes with a 2-hour glucose tolerance test or a hemoglobin A1C as a screen for comorbid conditions.
            - A fasting lipid panel is also helpful in assessing for comorbid conditions (hyperlipidemia)
        - Imaging in PMOS
            - In one or both ovaries
                a. 12 or more follicles measuring 2-9mm in diameter
                b. Increased ovarian volume (>10cm^3)
            - Evaluate endometrial thickness: estrogen exposure chronically without progesterone/progesterone withdrawal can cause endometrial thickening and possibly hyperplasia over time
            - Excessive intra-ovarian androgen causes granulosa cell dysfunction
                a. Oligo-ovulation
                b. Polycystic morphology: many small follicles with premature luteinization --> arrest of follicle maturation, no dominant follicle
        - Impacts of PMOS
            1. Menstrual irregularity
            2. Acne
            3. Hirsutism
            4. Insulin resistance/diabetes, weight gain
            5. Cardiovascular risk factors
            6. Infertility
            7. Increased risk of miscarriage
            8. Uterine Pathology- endometrial hyperplasia

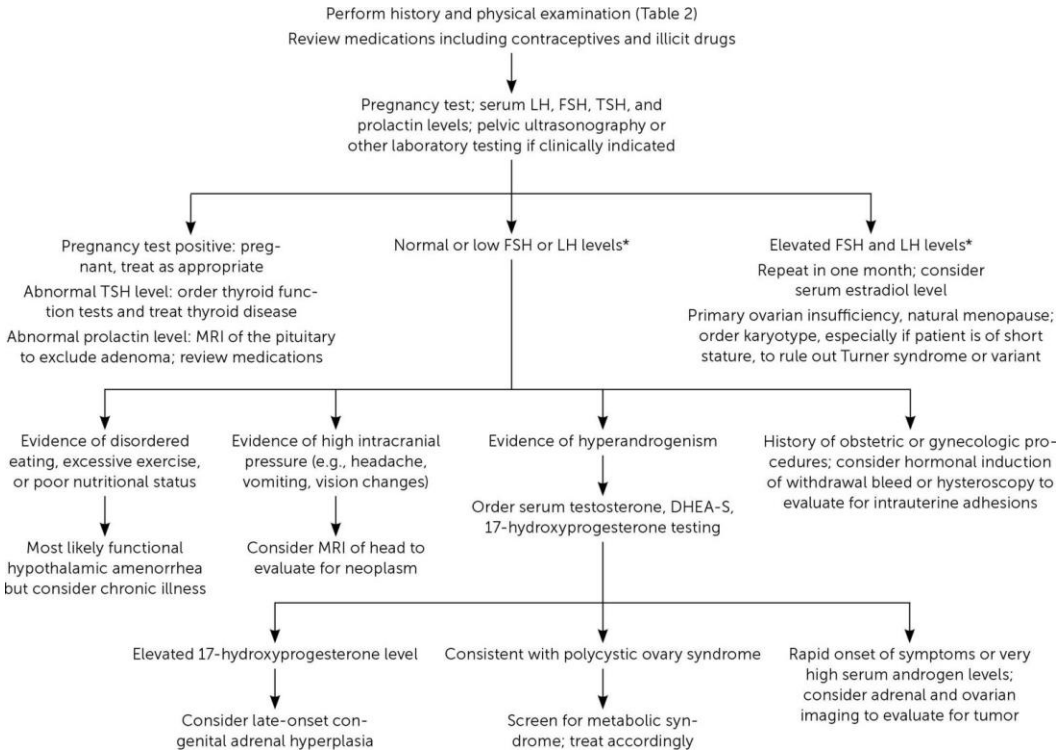
9. Anxiety/depression
- Other hyperandrogenism
  - Non-classic/late onset congenital adrenal hyperplasia (21-hydroxylase deficiency)
    1. Elevated 17-OHP
    2. Milder enzyme deficiency than classic CAH
  - Androgen secreting tumors
    1. May cause anovulation or endometrial atrophy
  - Exogenous androgens

5. Thyroid
   - Hypothyroidism: 10-35% will have menstrual irregularities depending on severity of thyroid dysfunction
   - Hyperthyroidism: 1-4% will have menstrual irregularities depending on severity of thyroid dysfunction
   - Can be heavy bleeding, abnormally light bleeding, or amenorrhea
   - Primary hypothyroidism (thyroid problem) can cause pituitary cell (lactotroph and thyrotroph) hyperplasia and thus elevated prolactin in the absence of lactotroph adenoma
6. Pituitary
   - Hyperprolactinemia: Elevated prolactin suppresses GnRH secretion leading to amenorrhea
     - Lactotroph (prolactin-secreting) adenomas --> hyperprolactinemia
     - Can also be elevated by primary hypothyroidism, lactation, stress, and recent intercourse
     - Medications can also increase prolactin, including CCBs and Methyldopa, as well as psych meds such as haloperidol and risperdal
     - Should be checked twice and other causes ruled out (thyroid, medication) before pituitary imaging ordered, especially if borderline elevation
   - Other sellar masses can cause deficient gonadotroph secretion
   - Other pituitary diseases: necrosis/infarction, radiation, infiltrative lesions
7. Hypothalamic
   - Functional hypothalamic amenorrhea
     - Decreased GnRH secretion --> decreased gonadotroph pulses/LH surge --> lack of ovulation
     - Eating disorders/very restricted fat intake, excessive exercise, high stress event or acute illness (typically transient)
     - Low estrogen levels can lead to bone loss over time
   - Hypothalamic tumors and infiltrative diseases
   - Systemic illness (T1DM, celiac disease)

Secondary Amenorrhea Work-up (H&P, imaging, laboratory)

Findings in the Evaluation of Amenorrhea

Findings	Associations
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Striae, buffalo hump, central obesity, hypertension	Cushing syndrome



Secondary Amenorrhea Treatment

- Secondary amenorrhea is most often a symptom, not a disease
    - Treat the underlying issue
        - Chronic disease
        - Eating disorder
        - Thyroid disease
        - Obesity/metabolic syndrome
    - Consider other health implications
        - If due to anovulation, consider endometrial protection and screen for comorbidities (esp in the setting of PMOS)
        - If due to inadequate estrogen production:
            a. Correct underlying problem, if possible
            b. Assess effects on other organs
                - Breast
                - Uterus
                - Bone
                - Cardiovascular health
                - Neurocognitive development
    - Periods themselves are not always the goal. What are your patient's concerns/goals?
        - Pregnancy?
        - Regular menses?
        - Avoiding "hormones"?
        - Improving health concerns (diabetes, hyperlipidemia)?
        - Fixing acne and/or hirsutism?

Answers to Study Questions

1) A 15yo patient presents to her pediatrician with her parent to discuss not having had a first menstrual period. She denies medical or surgical history and takes no medications. She denies any pelvic pain. She reports growth and development otherwise similar to her peers. Her vital signs are normal and her BMI is 23. Her physical exam reveals typical breast and pubic hair development. On pelvic exam, she is found to have a shortened vagina with no visible or palpable cervix. You are unable to palpate a uterus on exam, but it is difficult to fully evaluate due to patient discomfort. What is currently on your differential and what would your next steps in evaluation be?

Answer: The current differential includes the structural and outflow tract differences, including imperforate hymen, transverse vaginal septum, 5-alpha reductase deficiency, complete androgen insensitivity syndrome (AIS), and mullerian agenesis. The next step would be pelvic ultrasound to confirm the presence or absence of a uterus. If a normal uterus is present, the patient has either imperforate hymen or transverse vaginal septum, which can be distinguished on clinical exam by location in the vagina and thickness of the tissue. If a uterus is not present, the differential includes AIS and mullerian agenesis, which can be definitively distinguished by karyotype and serum testosterone levels.

2) A 27yo G1P0010 presents to her gynecologist after not having a period for 4 months. Prior to this, her periods had been regular and monthly. She denies past medical history. Her surgical history is significant for an appendectomy at age 12yo. She takes ibuprofen as needed for headaches, but no prescribed medications. She reports menarche at age 12y, initially irregular for the first 3-4 years, then regular for the last 10+ years. This is the longest she has ever gone without a period. She reports weight gain of about 10-15 pounds over the last year. She denies change in activity level or diet. She does note increasing fatigue and that she has had some worsening acne for the last two years. She does report constipation, but denies nipple discharge, hot flashes or vaginal concerns. She is sexually active with a male partner and uses condoms almost every time. Her VS are normal, BMI is 29, and general PE and pelvic exam are unremarkable. What is on your differential diagnosis and what are your next steps in assessment?

Answer: Your differential should include pregnancy, thyroid disease, PMOS, premature ovarian failure, and hyperprolactinemia. Based on history and physical exam, the most likely would be PMOS, thyroid disease, and pregnancy. The next step for this patient would be bloodwork, including thyroid testing, prolactin level, HCG level, androgen levels, 17-hydroxyprogesterone, and FSH/LH levels.